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On generalized Nash games and variational inequalities

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Abstract

We show that for a large class of problems a generalized Nash equilibrium can be calculated by solving a variational inequality. We analyze what solutions are found by this reduction procedure and hint at possible applications.
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1. Introduction

The standard definition of a noncooperative game in normal form usually requires that each player have a feasible set that is independent of the rivals' strategies. However, it was well understood from the early developments in the field, see e.g. [1,12,15], that in many cases the interaction between the players can take place (also) at the feasible set level. If one assumes that each player's feasible set can depend on the rival players' strategies we speak of generalized Nash equilibrium problems (GNEPs in the sequel).

GNEPs have received an increasing amount of attention in recent times because GNEPs naturally arise in the modeling of complex and important economical systems.

Let us give now a formal definition of the problem. There are N players, and each player ν controls the variables $x^\nu \in \mathbb{R}^{n_\nu}$. We denote by $\boldsymbol{x}$ the vector formed by all these decision variables: $\boldsymbol{x} \equiv ((x^1)^{\mathrm{T}}, (x^2)^{\mathrm{T}}, \ldots, (x^N)^{\mathrm{T}})^{\mathrm{T}}$, and by $\boldsymbol{x}^{-\nu}$ the vector formed by all the players' decision variables except those of player ν. To emphasize the ν-th player's variables within $\boldsymbol{x}$ we sometimes write $(x^\nu, \boldsymbol{x}^{-\nu})$ instead of $\boldsymbol{x}$.

The strategy of player ν must belong to a set $X_\nu(\boldsymbol{x}^{-\nu}) \subseteq \mathbb{R}^{n_\nu}$ that depends on the rival players' strategies. The aim of player ν, given the other players' strategies $\boldsymbol{x}^{-\nu}$, is to choose a strategy x^ν that solves the minimization problem

$$\begin{array}{ll} \text{minimize}_{x^\nu} & \theta_\nu(x^\nu, \boldsymbol{x}^{-\nu}) \\ \text{subject to} & x^\nu \in X_\nu(\boldsymbol{x}^{-\nu}). \end{array} \tag{1}$$

For any $\boldsymbol{x}^{-\nu}$, the solution set of this problem is denoted by $\mathcal{S}_\nu(\boldsymbol{x}^{-\nu})$. The GNEP is the problem of finding a

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vector $\bar{\boldsymbol{x}}$ such that

$$\bar{x}^\nu \in \mathscr{S}_\nu(\bar{\boldsymbol{x}}^{-\nu}) \quad \text{for all } \nu.$$

Such a point $\bar{\boldsymbol{x}}$ is called a (generalized) Nash equilibrium or, more simply, a solution of the GNEP. As usual in the literature, in this paper we make the following blanket assumptions:

(a) for every player ν, θ_ν is continuously differentiable in $\boldsymbol{x}$;
(b) for every player ν, the function $\theta_\nu(\cdot, \boldsymbol{x}^{-\nu})$ is pseudo-convex in x^ν.

An important case of GNEP is that explicitly introduced by Rosen in his fundamental paper [15]. In Rosen's setting a convex, compact set $\boldsymbol{X} \subset \mathbb{R}^n$ is given ($n = n_1 + \cdots + n_N$) and, for each $\boldsymbol{x}^{-\nu}$, the feasible set of the player ν is defined by

$$X_\nu(\boldsymbol{x}^{-\nu}) := \{x^\nu : (x^\nu, \boldsymbol{x}^{-\nu}) \in \boldsymbol{X}\}. \tag{2}$$

This situation occurs often in practice, most notably when the players share some resource available in limited amount (e.g. the capacity of a transmission or shipping channel). In fact, in practical instances of GNEPs, the feasible sets of the players are usually defined by (inequality) constraints. Rosen's setting corresponds to the fact that those constraints that depend on the other players' variables are shared by all the players and are convex with respect to all the variables. In the literature this setting is usually referred to as the "coupled constraints" case, after the paper [15] (where, however, $\theta_\nu(\cdot, \boldsymbol{x}^{-\nu})$ is assumed to be convex).

The study of GNEPs presents severe analytical difficulties, especially from the computational point of view. In the seminal paper [9], Harker singled out a class of GNEPs that can be solved by finding a solution of a variational inequality (VI); the latter being a much more tractable problem for which a very substantial set of results is available, see [8] for a review. Here we only recall that given a closed convex set $K \subseteq \mathbb{R}^n$ and a continuous function $G : K \to \mathbb{R}^n$, solving the VI defined by K and G (which is denoted by VI (K, G)) means finding a vector $x \in K$ such that

$$G(x)^{\mathrm{T}}(y - x) \geqslant 0 \quad \text{for all } y \in K.$$

The main aim of our paper is to show that Rosen's class of GNEPs can be solved by finding a solution of a single VI, thus extending to this very significant class of problems all the benefits of the VI reduction illustrated in [9]. Actually, we believe our approach is more direct and of wider applicability than that in [9], which, as we discuss in Section 4, also contains some inexactnesses.

2. Reduction to a VI

Define the function $\boldsymbol{F} : \mathbb{R}^n \to \mathbb{R}^n$ by

$$\boldsymbol{F}(\boldsymbol{x}) \equiv \begin{pmatrix} \nabla_{x^1}\theta_1(\boldsymbol{x}) \\ \vdots \\ \nabla_{x^N}\theta_N(\boldsymbol{x}) \end{pmatrix}. \tag{3}$$

Our main, simple result is given in following theorem.

Theorem 2.1. *Suppose that the GNEP satisfies assumptions* (a) *and* (b) *in the Introduction and assume further that the sets* $X_\nu(\boldsymbol{x}^{-\nu})$ *are defined by* (2) *with* $\boldsymbol{X}$ *closed and convex. Then, every solution of the VI* $(\boldsymbol{X}, \boldsymbol{F})$ *is a solution of the GNEP.*

Proof. Suppose that $\bar{\boldsymbol{x}}$ is a solution of the VI $(\boldsymbol{X}, \boldsymbol{F})$. We want to show that for every player ν, the corresponding $\bar{x}^\nu$ is a solution of (1). In order to see this let x^ν be any point in $X_\nu(\bar{\boldsymbol{x}}^{-\nu})$. It is clear that the point

$$\boldsymbol{y} \equiv \begin{pmatrix} \bar{x}^1 \\ \vdots \\ \bar{x}^{\nu-1} \\ x^\nu \\ \bar{x}^{\nu+1} \\ \vdots \\ \bar{x}^N \end{pmatrix}$$

belongs to $\boldsymbol{X}$. But since $\bar{\boldsymbol{x}}$ is a solution of the VI $(\boldsymbol{X}, \boldsymbol{F})$ we can write

$$0 \leqslant \boldsymbol{F}(\bar{\boldsymbol{x}})^{\mathrm{T}}(\boldsymbol{y} - \bar{\boldsymbol{x}}) = \nabla_{x^\nu}\theta_\nu(\bar{x}^\nu, \bar{\boldsymbol{x}}^{-\nu})^{\mathrm{T}}(x^\nu - \bar{x}^\nu),$$

which, under the assumed pseudo-convexity of θ_ν, shows, by the minimum principle, that $\bar{x}^\nu$ is a solution of (1). □

We remark that the theorem above does not state that any solution of the GNEP is also a solution of VI $(\boldsymbol{X}, \boldsymbol{F})$ and in fact this is not true in general (see also Section 5 for an example). Therefore, it could happen

that the GNEP has a solution, but VI $(\boldsymbol{X}, \boldsymbol{F})$ has none. If the aim of introducing VI $(\boldsymbol{X}, \boldsymbol{F})$ is that of finding a solution of the GNEP by solving VI $(\boldsymbol{X}, \boldsymbol{F})$, we must be sure that this VI has at least one solution. Any standard condition for the existence of a solution of a VI could be applied in principle. The following proposition highlights a very simple, but practical condition that ensures the existence of a solution to VI $(\boldsymbol{X}, \boldsymbol{F})$ (see for example [8]).

Proposition 2.2. *Assume the setting of Theorem* 2.1 *and suppose, additionally, that the set* $\boldsymbol{X}$ *is compact. Then a solution of VI* $(\boldsymbol{X}, \boldsymbol{F})$ *exists and this is also a solution of the corresponding GNEP.*

We stress that this is just one possible condition for the existence of a solution to VI $(\boldsymbol{X}, \boldsymbol{F})$.

Remark 2.1. After this paper was completed Christian Kanzow kindly brought to our attention that a result very similar to Theorem 2.1 is given as [2, Proposition 4, Chapter 9]. Theorem 2.1 is quite simple and we are sure it has been "discovered" several times by many independent researchers. Nevertheless, the equivalence between the GNEP and a VI under the setting of the present paper seems to have eluded the attention of most practitioners and also of many more mathematically oriented researchers. We believe it is useful to have Theorem 2.1 here, especially in view of the simple proof we give.

3. Solution selection

Having understood that in passing from GNEP to VI not all the GNEP solutions are preserved, the obvious next question to ask is what kind of solution are obtained by solving the VI. The answer is surprisingly neat and elegant; to give it we first need to specify better a structure for the set $\boldsymbol{X}$ and introduce the KKT conditions for both the GNEP and the VI. To this end we suppose that $\boldsymbol{X}$ is defined by a finite number of constraints:

$$\boldsymbol{X} \equiv \{\boldsymbol{x} \in \mathbb{R}^n : g(\boldsymbol{x}) \leqslant 0\}, \tag{4}$$

where $g : \mathbb{R}^n \to \mathbb{R}^m$ and we suppose that all the constraints $g_i : \mathbb{R}^n \to \mathbb{R}$, $i = 1, \ldots, m$ are convex and continuously differentiable. Note that usually there are two groups of constraints, those which depend on $\boldsymbol{x}$ and those depending on a single player's variables x^ν. For notational simplicity, and following the original paper by Rosen [15], we include these latter constraints in the former ones.

Suppose that $\boldsymbol{x}$ is a solution of the GNEP. Then, if for player ν a suitable constraint qualification holds, there is a vector $\lambda^\nu \in \mathbb{R}^m$ of multipliers so that the classical KKT conditions

$$\begin{aligned} &\nabla_{x^\nu}\theta_\nu(x^\nu, \boldsymbol{x}^{-\nu}) + \nabla_{x^\nu} g(x^\nu, \boldsymbol{x}^{-\nu})\lambda^\nu = 0, \\ &0 \leqslant \lambda^\nu \perp g(x^\nu, \boldsymbol{x}^{-\nu}) \leqslant 0 \end{aligned} \tag{5}$$

are satisfied. Note that if a constraint g_i is not active at $\boldsymbol{x}$ then necessarily the corresponding multiplier λ_i^ν is zero for all players. On the other hand, multipliers corresponding to the same active constraint can obviously be different for each player. We now show that the solutions of the GNEP that are preserved in passing to the VI are exactly those for which all players have the same multipliers for all constraints.

Consider the KKT conditions for the VI $(\boldsymbol{X}, \boldsymbol{F})$. Assuming that a constraint qualification holds at a solution $\boldsymbol{x}$, the KKT conditions (see [8]) read as

$$\boldsymbol{F}(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} g(\boldsymbol{x})\lambda = 0,$$

$$0 \leqslant \lambda \perp g(\boldsymbol{x}) \leqslant 0.$$

Taking into account definition (3) of $\boldsymbol{F}$, these conditions can be written as

$$\begin{aligned} &\begin{pmatrix} \nabla_{x^1}\theta_1(\boldsymbol{x}) \\ \vdots \\ \nabla_{x^N}\theta_N(\boldsymbol{x}) \end{pmatrix} + \begin{pmatrix} \nabla_{x^1} g(\boldsymbol{x}) \\ \vdots \\ \nabla_{x^N} g(\boldsymbol{x}) \end{pmatrix} \lambda = 0, \\ &0 \leqslant \lambda \perp g(\boldsymbol{x}) \leqslant 0. \end{aligned} \tag{6}$$

We are now ready to prove the result we mentioned at the beginning of this section.

Theorem 3.1. *Suppose that the GNEP satisfies assumptions* (a) *and* (b) *in the Introduction and assume further that the sets* $X_\nu(\boldsymbol{x}^{-\nu})$ *are defined by* (2) *with* $\boldsymbol{X}$ *given by* (4) *and convex.*

(i) *Let* $\boldsymbol{x}$ *be a solution of VI* $(\boldsymbol{X}, \boldsymbol{F})$ *at which the KKT conditions* (6) *hold. Then* $\boldsymbol{x}$ *is a solution of the*

GNEP at which the KKT conditions (5) *hold with* $\lambda^1 = \lambda^2 = \cdots = \lambda^N = \lambda$.

(ii) *Viceversa, let* $\boldsymbol{x}$ *be a solution of the GNEP at which KKT conditions* (5) *hold with* $\lambda^1 = \lambda^2 = \cdots = \lambda^N$. *Then* $\boldsymbol{x}$ *is a solution of VI* $(\boldsymbol{X}, \boldsymbol{F})$.

Proof. (i) If (6) holds, a direct comparison with (5) shows that $\boldsymbol{x}$ satisfies the KKT conditions (5) with $\lambda^1 = \lambda^2 = \cdots = \lambda^N = \lambda$. By assumption (b) and the convexity of $\boldsymbol{X}$ these latter KKT conditions are sufficient to guarantee that $\boldsymbol{x}$ is a solution of the GNEP.

(ii) If $\boldsymbol{x}$ is a solution of the GNEP at which the KKT conditions (5) hold with $\lambda^1 = \lambda^2 = \cdots = \lambda^N$, the KKT conditions (6) are obviously satisfied (with $\lambda = \lambda^1$); but it is well known (see e.g. [8]) that the satisfaction of these latter conditions ensures that $\boldsymbol{x}$ is a solution of VI $(\boldsymbol{X}, \boldsymbol{F})$. □

Remark 3.1. If we assume that $\boldsymbol{X}$ is bounded and Slater constraint qualification holds, Proposition 2.2, the constraint qualification and the theorem above show that a solution of the GNEP exists at which the multipliers of all players are the same. This was proved already by Rosen [15] in 1965. He called such a solution a "normalized equilibrium point". We believe that our proof gives a different and useful insight on the nature of "normalized equilibria". This modern point of view on the subject was not possible back in 1965 when the theory of VI was still in its infancy.

4. Comparison to Harker's approach

In [9], Harker laid the foundations of the (quasi) variational inequality approach to generalized Nash games. Even if the fact that a GNEP can be reformulated (under suitable convexity assumptions) as a QVI was well known since Bensoussan's paper [4] in 1974, it was Harker who clearly showed the relevance of the reduction to a QVI also from the computational point of view and investigated various important consequences of this reduction. Furthermore, it was Harker who pointed out the practical significance of the case in which the GNEP can be reduced to a VI. This latter issue is certainly fundamental, and it is safe to say that to date the only GNEPs for which one can reasonably speak of solution procedures are those that can be reduced to a VI. From this point of view we believe we give a significant contribution, the more so since, as we pointed out in the Introduction, there seem to be some imprecisions in the treatment of this issue in [9]. Consider the GNEP and define the point-to-set mapping

$$\Omega(\boldsymbol{x}) = X_1(\boldsymbol{x}^{-1}) \times \cdots \times X_N(\boldsymbol{x}^{-N}).$$

Harker showed that any solution of the VI $(\Gamma, \boldsymbol{F})$ (with $\boldsymbol{F}$ given by (3)) is a solution of the GNEP if Γ is a nonempty, compact, convex set in $\mathbb{R}^n$ such that the following two conditions are satisfied:

(i) $\Omega(\boldsymbol{x}) \subseteq \Gamma$ for all $\boldsymbol{x} \in \Gamma$;
(ii) $\boldsymbol{x} \in \Omega(\boldsymbol{x})$ for all $\boldsymbol{x} \in \Gamma$.

The problem with this result is, we believe, that it is not simple to give classes of problems for which (i) and (ii) are satisfied for some easily calculated Γ. Rosen's setting analyzed in this paper is certainly not covered (when $\Gamma = \boldsymbol{X}$, as suggested in [9] for polyhedral $\boldsymbol{X}$). In fact, in Rosen's setting, condition (ii) is obviously always satisfied, but condition (i) need not, except when $\boldsymbol{X}$ is a rectangle. We illustrate this with a simple polyhedral example. Suppose that

$$\boldsymbol{X} = \{(x^1, x^2) \in \mathbb{R}^2 : x^1 + x^2 \leqslant 2,\ x^2 - x^1 \geqslant 0,\ x^1, x^2 \geqslant 0\}.$$

We see that $\boldsymbol{X}$ is the triangle with vertices (0, 0), (1, 1), and (0, 2). Take $\Gamma = \boldsymbol{X}$ and consider the point $(0, 1) \in \Gamma$. We have

$$X_1(1) = [0, 1], \quad X_2(0) = [0, 2],$$
$$\Omega(0, 1) = [0, 1] \times [0, 2].$$

It is easily seen that $\Omega(0, 1)$ is a rectangle that properly contains the triangle $\Gamma = \boldsymbol{X}$, so that (i) is not satisfied. Another example of this phenomenon is given in the next section. Note that in [9] the only exhibited class of problems for which conditions (i) and (ii) are claimed to be satisfied is (using our terminology) the case of Rosen's setting with $\boldsymbol{X}$ polyhedral. However, we just showed that this is not true. The trouble in [9] seems to be in the unsubstantiated claim that "assumption (i) is always met...".

Harker also discussed what kind of solution of the GNEP is obtained in passing from the original GNEP to a VI reformulation. Again our Theorem 3.1 moves along the lines he set, but, we believe, is simpler,

clearer and of wider applicability. Actually, we hope Theorem 3.1 finally gives firm ground and a clear meaning to the widely spread perception among researchers working in the field of GNEPs that the case in which "there exist common multipliers" is easy.

5. An example and final considerations

We illustrate the results obtained so far with an example. In order to make this example more readable we deviate from the notation adopted and indicate the variable of the first player with x and that of the second with y. Similarly, the multipliers of the first player are λ and those of the second μ. Consider the game with two players:

$$\begin{array}{ll} \min_x & (x-1)^2 \\ & x+y\leqslant 1 \end{array} \qquad \begin{array}{ll} \min_y & (y-\frac{1}{2})^2 \\ & x+y\leqslant 1. \end{array}$$

The optimal solution sets are given by

$$\mathscr{S}_1(y)=\begin{cases} 1 & \text{if } y\leqslant 0, \\ 1-y & \text{if } y\geqslant 0, \end{cases}$$

$$\mathscr{S}_2(x)=\begin{cases} \frac{1}{2} & \text{if } x<\frac{1}{2}, \\ 1-x & \text{if } x\geqslant \frac{1}{2}. \end{cases}$$

A point (x, y) is a solution if

$$\begin{pmatrix} x \\ y \end{pmatrix} \in \begin{pmatrix} \mathscr{S}_1(y) \\ \mathscr{S}_2(x) \end{pmatrix}.$$

Then it is easy to check that the solutions of this problem are given by $(\alpha, 1-\alpha)$ for every $\alpha \in [\frac{1}{2}, 1]$. Therefore this problem has infinitely many solutions. Since at each solution the linear independence constraint qualification holds, for each such solution there are unique multipliers $\lambda(\alpha)$, for the first player, and $\mu(\alpha)$, for the second player, that together with $(\alpha, 1-\alpha)$ satisfy the KKT conditions. If we set the gradients of the Lagrangians of the two players' minimization problems to zero, we deduce that

$$\lambda(\alpha)=2-2\alpha, \quad \mu(\alpha)=2\alpha-1.$$

Therefore, there is only one solution for which $\lambda(\alpha)=\mu(\alpha)$, and this is obtained for $\alpha=\frac{3}{4}$. Corresponding to this value we thus obtain

$$(\bar{x}, \bar{y})=(\tfrac{3}{4}, \tfrac{1}{4}), \quad \bar{\lambda}=\tfrac{1}{2}=\bar{\mu}.$$

Consider now the VI $(\boldsymbol{X}, \boldsymbol{F})$ that, in this case, is obtained by setting

$$\boldsymbol{X}=\{(x, y)\in\mathbb{R}^2 : x+y\leqslant 1\}, \quad \boldsymbol{F}=\begin{pmatrix} 2x-2 \\ 2y-1 \end{pmatrix}.$$

$\boldsymbol{F}$ is clearly strongly monotone and therefore this VI has a unique solution which is given by $(\frac{3}{4}, \frac{1}{4})$ as can be checked by using the definition of VI (note that in this example the set $\boldsymbol{X}$ is not compact, but since $\boldsymbol{F}$ is strongly monotone existence and uniqueness of the solution are guaranteed). Furthermore, if we write down the KKT conditions for this VI, we can check that the multiplier corresponding to the unique constraint that defines $\boldsymbol{X}$ is $\frac{1}{2}$ (i.e. the common value of the multipliers of the generalized Nash game in the solution $(\frac{3}{4}, \frac{1}{4})$). We see then that while the original generalized Nash game has infinitely many solutions, VI $(\boldsymbol{X}, \boldsymbol{F})$ has only one solution, more precisely, the one solution of the generalized game for which the shared constraint has equal multipliers.

We can also use this example to illustrate again the fact that condition (i) in the previous section need not be satisfied. Consider the point $(0, 0)$, that clearly belongs to $\boldsymbol{X}$. Using the notation of the previous section, in this case it is immediate to verify that $\Omega(0, 0)$ is the rectangle $[-\infty, 1]\times[-\infty, 1]$ which is not contained in $\boldsymbol{X}$ (just consider point $(1, 1)$, for example).

We conclude by noting that although very simple the results of this paper have several applications. In particular, applications in the development of a Newton method for the solution of the GNEP are analyzed in detail in [7]. However, we believe that it is in the development of globally convergent algorithms for the calculation of a solution to a GNEP that the results we presented here can have noticeable consequences. We briefly illustrate this. As far as we are aware of, there are only few globally convergent algorithms for the solution of GNEPs. Basically, if one restricts to algorithms that are guaranteed to find a solution by performing well defined tasks and by solving subproblems that are surely solvable, they boil down to two: the relaxation algorithm (see for example [3,5,6,10,11,14,16]) and the reduction to an MPEC problem (see [13] and references therein). In both these approaches Rosen's setting is assumed to begin with. Furthermore, some other (strong) assumptions are made that imply that the mapping $\boldsymbol{F}$ is at least

strictly monotone. Note that in general it may look unreasonable to expect $\boldsymbol{F}$ to be strictly monotone, since the concatenation of partial gradients of convex functions need not be monotone, let alone strictly monotone. However, in many applications of GNEPs strict monotonicity of $\boldsymbol{F}$ can be expected to hold since it basically amounts to saying that each player has more control on his/hers own objective function than the other players do. By using the reduction to VI $(\boldsymbol{X}, \boldsymbol{F})$ we can use a host of algorithms (old and new) that are guaranteed to find a solution (if one exists) by making much simpler assumptions on the problem. For example, if one uses a hyperplane projection method, convergence can be guaranteed by merely assuming that $\boldsymbol{F}$ is pseudo-monotone with respect to the solution set (see [8, Chapter 12]). We leave the detailed study of these issues to future research.

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